Real-World Application

1 Problem Background

The Rhône-Alpes region in southeastern France includes a diverse geographical landscape, ranging from densely populated urban areas such as Lyon, Grenoble, and Saint-Étienne to mountainous and rural areas in the French Alps.

Providing reliable wireless communication coverage across the entire region is challenging due to geographical constraints, population distribution, and the cost of deploying communication infrastructure.

A practical approach is to identify a set of potential locations for wireless communication towers and select the smallest number of towers that can collectively provide coverage to all target areas.

This problem can be formulated as a **Minimum Set Cover Problem**, where each candidate tower covers a subset of geographical areas, and the objective is to minimize the total number of selected towers while ensuring that every target area is covered.

2 Problem Definition

The objective is to determine the minimum number of wireless communication towers required to cover all target areas in the Rhône-Alpes region.

For modeling purposes, the region is divided into a finite set of geographical coverage units. Each unit can represent a municipality, a grid cell, or a population center. A set of candidate tower locations is then identified based on suitable deployment sites, such as existing infrastructure, elevated terrain, or accessible land.

A tower is assumed to provide coverage to a geographical unit if the distance between the tower and that unit is within a predefined coverage radius. In this simplified model, terrain obstruction and signal interference are not considered.

The problem is formulated as follows:

- $U = \{u_1, u_2, \ldots, u_n\}$ is the universe of target geographical areas that require wireless coverage.
- $S = \{S_1, S_2, \ldots, S_m\}$ is the collection of candidate tower coverage sets.
- Each $S_j \subseteq U$ represents the geographical areas covered by candidate tower j.
- $x_j \in \{0, 1\}$ indicates whether a tower is selected at candidate location j.

The optimization problem is:

$$\min \sum_{j=1}^{m} x_j \tag{1}$$

Subject to:

$$\sum_{j:u_i \in S_j} x_j \geq 1, \quad \forall u_i \in U \tag{2}$$

$$x_j \in \{0,1\}, \quad j = 1, \ldots, m \tag{3}$$

The objective minimizes the number of deployed towers, while the constraints ensure that every target geographical area is covered by at least one selected tower.

3 Illustrative Example

Consider a simplified scenario involving 10 target geographical areas and 6 potential tower locations. Each candidate tower can cover multiple target areas depending on its location and assumed coverage radius.

The coverage sets are defined as follows:

- Tower 1: $\{A, B, C, D\}$
- Tower 2: $\{C, D, E, F\}$
- Tower 3: $\{E, F, G\}$
- Tower 4: $\{G, H, I\}$
- Tower 5: $\{A, H, J\}$
- Tower 6: $\{B, D, I, J\}$

The goal is to select the smallest subset of candidate towers whose combined coverage includes all 10 geographical areas.

This example is illustrative and does not represent actual tower coverage in the Rhône-Alpes region.

4 Data Preparation and Instance Generation

Since actual telecommunications infrastructure data may be difficult to obtain, the project can use a combination of geographical data and simulated coverage information.

The following approach can be adopted:

1. **Define target areas:** Divide the Rhône-Alpes geographical area into grid cells or select municipalities as coverage units.
2. **Generate candidate locations:** Identify potential tower locations distributed across the region.

3. **Determine coverage:** Calculate the distance between each candidate tower and each target area. A target area is considered covered if its distance from the candidate tower does not exceed the predefined coverage radius.

4. **Construct coverage sets:** For each candidate location, create a set containing all geographical areas within its coverage radius.

5. **Generate test instances:** Vary the number of target areas, candidate tower locations, and coverage radius to produce instances with different sizes and coverage densities.

For a more realistic implementation, geographical coordinates can be obtained from OpenStreetMap or other publicly available geographical datasets. The simplified distance-based coverage model can later be extended to account for terrain elevation and line-of-sight constraints.

5 Application of Minimum Set Cover Algorithms

The algorithms implemented in this project can be applied to the generated instances:

- **Brute Force:** Enumerate possible combinations of tower locations to find the optimal solution for small instances.
- **Branch and Bound:** Explore candidate combinations while pruning branches that cannot improve the current best solution.
- **Greedy:** Iteratively select the tower that covers the largest number of currently uncovered areas.
- **Dynamic Programming:** Compute an optimal solution using states representing covered subsets of geographical areas, for sufficiently small instances.
- **Additional methods and heuristics:** Explore alternative strategies to obtain good solutions for larger instances.

All algorithms use the same coverage-set representation to ensure a fair comparison.

5.1 Evaluation Metrics

The algorithms will be evaluated using the following metrics:

- **Number of selected towers:** Measures the size of the resulting deployment.
- **Coverage completeness:** Verifies whether all target geographical areas are covered.
- **Running time:** Measures the computational efficiency of each algorithm.
- **Solution quality:** Compares the obtained solution against the optimal or best-known solution.

- **Scalability:** Evaluates algorithm performance as the number of target areas and candidate locations increases.

For small instances, the optimal solution obtained through Brute Force or Branch and Bound can serve as a reference. For larger instances, the best-known solution can be used for comparison.

6 Expected Outcomes and Practical Significance

This application demonstrates how the Minimum Set Cover Problem can support infrastructure planning by identifying a small number of communication tower locations that collectively cover all target areas.

The experimental results can help answer several practical questions:

- How many towers are required to achieve complete coverage under different coverage-radius assumptions?
- How does increasing the number of candidate locations affect solution quality?
- Which algorithm provides the best trade-off between solution quality and computational time?
- How well do approximation and heuristic algorithms perform on large geographical instances?

The results could provide a preliminary decision-support model for wireless infrastructure planning, particularly in regions with a combination of urban, rural, and mountainous geographical characteristics.

7 Scope and Limitations

This project focuses on the simplified, unweighted Minimum Set Cover Problem. All candidate towers are assumed to have equal deployment costs and identical coverage characteristics, except for their geographical locations.

The model does not explicitly account for:

- Terrain obstruction and signal propagation.
- Differences in tower installation and maintenance costs.
- Population density and varying coverage demand.
- Signal interference and overlapping network capacity.

These factors could be incorporated in future work by extending the model to Weighted Set Cover or more realistic wireless network optimization problems.

Finally, Rhône-Alpes refers to the former administrative region that became part of the larger Auvergne-Rhône-Alpes region following the 2016 territorial reform. For this project, the geographical scope is limited to the former Rhône-Alpes area.